Highlights

TradeSage: Unraveling Markets with AI

Nashit Budhwani Fahad Zahid Lohit Ashwa Muhammad Rafi

- Introduction
- Related Work
- Theoretical Foundations
- Methodology
- Conclusion
- Bibliography
- My Appendix
- CRediT authorship contribution statement

TradeSage: Unraveling Markets with AI⋆,⋆⋆

Mr Nashit Budhwani[a,c,∗,1] (Researcher), Mr Fahad Zahid[a,d] (Researcher), Mr Lohit Ashwa[a,e] (Researcher), Dr Muhmmad Rafi[a,f] (Supervisor)

[a]*National University of Computer and Emerging Sciences - FAST, St-4, Sector 17-D, NH 5 , Karachi, 75078, Pakistan*

ARTICLE INFO

Keywords:
Large language models
Natural language processing
Trade data
Vector databases
Data analysis
Trade insights
Market understanding
Global commerce
Economic intelligence

ABSTRACT

Large language models (LLMs) have demonstrated potential to revolutionize natural language processing, generating enthusiasm in trade. The primary challenge lies in obtaining reliable trade data. TradeSage ventures into AI to develop an LLM application that transforms raw trade data into relatable narratives, enabling the model to grasp intricate trade patterns. Utilizing frameworks such as LangChain and vector databases, TradeSage enhances the analysis and communication of trade insights. By generating clear stories from complex datasets, TradeSage helps businesses, governments, and researchers better understand markets and trade flows, making information accessible to non-experts and facilitating smarter decisions in global commerce.

1. Introduction

1.1 Background

Trade, the backbone of global economies, is a dynamic and intricate system that propels economic growth, builds international connections, and shapes global markets. Timely and accurate information is essential for navigating this complex web of commerce. Traders, investors, and businesses involved in international trade or monitoring global markets rely on a steady stream of insights to make informed decisions.

1.2 The Potential of Large Language Models in Trade

Artificial intelligence, particularly Large Language Models (LLMs), is poised to revolutionize natural language processing across various fields, including trade. LLMs, such as those developed by Vaswani et al. (2017) and Brown et al. (2020), have shown remarkable capabilities in understanding and generating natural language text. However, the primary challenge lies in obtaining reliable and valuable trade data to train these models effectively.

1.3 TradeSage: Bridging AI and Trade

TradeSage is an innovative project that ventures into artificial intelligence to create a user-friendly application leveraging LLMs to transform raw trade data into understandable narratives. These narratives serve as training data, enabling the LLM to grasp intricate trade patterns and relationships. By utilizing frameworks such as LangChain and vector databases, TradeSage enhances the LLM's ability to analyze and communicate trade insights.

1.4 Local and Global Trade Analysis

TradeSage will cater to the needs of traders, investors, and businesses, particularly those operating in Pakistan's dynamic trading landscape. The application will integrate world trade analysis and local trade analysis to offer comprehensive and customized trade insights. World trade analysis involves examining international trade patterns, trends, and dynamics using diverse data sources, while local trade analysis focuses on the implications of international trade on specific regions, assessing the impact on local industries and communities.

1.5 Challenges in International Trade

International trade faces numerous challenges, including changing regulations, geopolitical tensions, and market volatility. Traditional methods of data analysis often struggle to keep pace with these dynamic factors. TradeSage acknowledges these challenges and aims to address them by leveraging advanced AI technology.

1.6 Objectives of the TradeSage Project

The primary objective of TradeSage is to develop an interactive system that utilizes Generative AI and fine-tuned LLMs to streamline trade analysis and decision-making processes. The project aims to democratize access to global trade insights by transforming vast repositories of trade data into actionable information. By doing so, TradeSage seeks to empower policymakers, businesses, researchers, and the general public to make informed decisions and gain deeper insights into global trade dynamics.

1.7 Significance of the TradeSage Project

The significance of TradeSage lies in its potential to redefine trade data analysis and communication. By providing clear and accessible narratives from complex data sets, TradeSage enhances understanding of market trends, trade flows, and economic conditions. This not only aids businesses, governments, and researchers but also makes trade information more accessible to non-experts, enabling smarter choices in global trade. Ultimately, TradeSage fuses technology and economic intelligence, promising to enable data-driven decision-making in international trade.

1.8 Vision for TradeSage

In order to address the requirements of effective trade data analysis, the project's vision emphasizes a data-oriented approach that prioritizes high-quality data through advanced cleaning and preprocessing techniques within an end-to-end application structured into four distinct layers: a data retrieval layer focusing on news, social media posts, and time-sensitive market data; a data engineering layer utilizing NLP techniques to handle challenges of time sensitivity and signal-to-noise ratios; an LLMs layer employing methodologies such as Retrieval-Augmented Generation (RAG) and fine-tuning to manage dynamic trade data; and a user-interface layer providing customized responses to user queries, showcasing the model's potential. This innovative solution leverages language models and data engineering to create a robust, user-friendly application capable of analyzing complex trade data, significantly contributing to advancements in international trade research and decision-making.

research and practices.

* This research was submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science.

** We extend our heartfelt gratitude to our supervisor, Dr. Muhammad Rafi, whose unwavering support and guidance propelled us toward excellence throughout the development of the TradeSage

*Corresponding author

**Corresponding author

***Corresponding author

ORCID(s):

2. RelATED WORK

1. **BloombergGPT:** BloombergGPT is a 50-billion-parameter LLM developed by Bloomberg, specifically tailored for the financial sector. This model was trained on an extensive dataset comprising 363 billion tokens from Bloomberg's proprietary financial data and an additional 345 billion tokens from general-purpose datasets, totaling over 700 billion tokens. The integration of both domain-specific and general data enables BloombergGPT to excel in financial tasks such as sentiment analysis, named entity recognition, news classification, and question answering, while maintaining robust performance on general NLP benchmarks. Notably, BloombergGPT outperforms existing open models of similar size on financial tasks without compromising on general language understanding capabilities. However, its proprietary nature limits accessibility for broader research and development purposes.

2. **FinBERT:** FinBERT is a specialized adaptation of the Bidirectional Encoder Representations from Transformers (BERT) architecture, fine-tuned for financial sentiment analysis. It was trained on a vast corpus of financial texts, including analyst reports, financial news, and corporate filings, enabling it to accurately discern sentiment in financial documents. FinBERT has demonstrated superior performance in tasks such as financial news classification, entity recognition, and sentiment prediction, making it a valuable tool for market trend analysis and investment decision-making. Its ability to understand the nuances of financial language allows it to assist in tasks like assessing market reactions to news and identifying potential investment opportunities.

3. **ManGPT:** Developed by Man Group, the world's largest publicly-listed hedge fund, ManGPT is an in-house LLM utilized to enhance trading strategies and operations. The firm has been integrating AI and machine learning tools for over a decade, with ManGPT playing a pivotal role in their technological advancements.

4. **Colosseum 355B:** The Database for Annotation, Visualization and Integrated Discovery (DAVID) offers a suite of functional annotation tools to interpret the biological significance of gene lists. It leverages the DAVID Knowledgebase, which consolidates functional anno- tations from multiple sources. DAVID tools enable users to identify enriched biological themes, discover functional-related gene groups, cluster redundant an- notation terms, visualize genes on pathway maps, ex- plore gene-disease associations, and perform various other analyses. This comprehensive platform empow- ers researchers to gain insights into the functional relevance of gene lists.

5. **PanGu-Σ:** Developed by Huawei, PanGu-Σ is a multimodal LLM designed for various industries, including finance. It features a hierarchical architecture that allows customization for specific tasks within the financial sector, such as risk assessment, portfolio optimization, and financial forecasting. PanGu-Σ's adaptability makes it a valuable tool for financial institutions aiming to integrate advanced AI capabilities into their operations.

3. Theoretical Foundations

1. Artificial Intelligence in Trading:

Understanding international trade, with its intricate data and constant changes, is a significant challenge. Large Language Models (LLMs) offer a promising solution, but using them effectively is key. TradeSage addresses this by developing an AI-driven LLM application designed to transform complex trade data into clear, comprehensible narratives. Rather than merely processing raw data, TradeSage trains the LLM to recognize nuanced relationships within trade information, integrating advanced frameworks like LangChain and vector databases to enhance analytical and communicative abilities. TradeSage's impact spans various sectors, including businesses, governments, and research institutions, delivering actionable insights from detailed datasets. Its distinguishing feature is a commitment to democratizing trade data, making it accessible to non-experts and empowering informed decision-making. This emphasis on accessibility marks a critical step toward a more inclusive and transparent international trade environment.

2. Current Technologies in Trade Analysis

Trade analysis relies on processing and interpreting vast amounts of data, but traditional methods often struggle with the complexities and dynamic nature of global trade. Existing solutions for trade-related queries are either limited in scope or lack the sophistication needed to handle diverse inquiries effectively. TradeSage addresses this gap by integrating Generative AI with a finely tuned Large Language Model (LLM), leveraging advanced AI techniques and data engineering to transform diverse datasets into practical, actionable insights, enhancing the accuracy and efficiency of trade analysis.

3. Current Technologies in Trade Analysis

Despite advancements, current language models face significant challenges in trade analysis, including limited focus on trade-specific language models, scarcity of domain-specific training data, insufficient integration of external knowledge, inadequate exploration of effective fine-tuning strategies, and limited understanding of user preferences in trade applications. TradeSage addresses these gaps by combining Generative AI and advanced fine-tuning methods, optimizing the integration of external data, and employing Reinforcement Learning from Human Feedback (RLHF) to deliver personalized, actionable insights. This approach enhances AI's ability to navigate the complexities of international trade, benefiting diverse stakeholders.

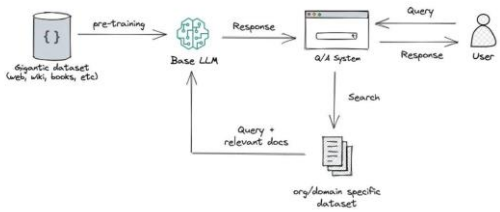
4. Review of Relevant Literature

Language modeling has evolved significantly since its inception, transitioning from early N-gram models to advanced transformer-based architectures like GPT-3 and GPT-4, which excel in diverse generative tasks. Recent research emphasizes domain-specific training, demonstrating superior results in biomedical and financial contexts with models like BioBERT, SciBERT, and BloombergGPT. Addressing the accessibility and customization challenges, TradeSage introduces an open-source Trade LLM leveraging Reinforcement Learning from Human Feedback (RLHF) and Retrieval Augmented Generation (RAG), aiming to deliver personalized, domain-specific trade assistance by enhancing general LLM capabilities within the trade sector.

4. Methodology

Our methodology began with raw trade data provided as a SQL Server database image by ABC Company, which initially lacked defined relationships; to capture the underlying meaning, we collaborated with the company's engineers to establish necessary links. In Phase I of data conversion, we used a LangChain dataframe agent powered by GPT-3 to derive row-wise textual descriptions from the table, followed by generating table-wise sentences in Phase II and constructing prompt completions using SQL queries in Phase III. Our approach combined effective prompt engineering techniques—including zero-shot, one-shot, and two-shot learning—with a Retrieval-Augmented Generation (RAG) system via the LangChain framework to integrate both structured and unstructured data sources, thereby enhancing response accuracy while mitigating issues like data leakage and hallucination. Further refinement involved fine-tuning the model through adaptive, behavioral, and parameter-efficient methods, with the workflow starting from comprehensive dataset curation, progressing from a paid RAG implementation with GPT-3 and Pinecone to an open-

source solution using Llama 2 via LlamaIndex, and culminating in a supervised fine-tuning process on 1500 instances, including model quantization to prevent GPU memory exhaustion. Although the fine-tuning yielded some improvements, it also indicated overfitting, underscoring the need for further optimization



5. Conclusion

5.1 Summary of Findings

In conclusion, this project aimed to develop an open-source solution capable of answering trade-related queries in easy-to-understand natural language narratives. Through meticulous methodology and experimentation, several key findings have been observed.

5.2 Performance of TradeSage

TradeSage effectively leveraged frameworks like Llama V2, V3, and vector indexer like Llama Index to transform raw trade data into comprehensible narratives. The models demonstrated commendable performance in analyzing and communicating trade insights, making complex data accessible to a broader audience, including non-experts.

5.3 Challenges Encountered

Despite the successes, the project faced significant challenges. One of the primary issues was hallucination, where the model produced inaccurate or fabricated information. Additionally, running the models efficiently required high-end GPUs, such as the A100, which cost around USD 8,000, posing a substantial financial barrier.

5.4 Fine-Tuning Difficulties

During the fine-tuning process, the model encountered issues with overfitting and underfitting. Striking the right balance between these two proved challenging, as the model either failed to generalize well or did not learn adequately from the training data.

5.5 Overall Achievement of Objectives

In summary, this project successfully developed an open-source trade analysis solution that transforms complex trade data into accessible narratives which is deployed at ABC Company's website. While the model showed promise in making trade information more understandable and democratizing access to trade insights, challenges such as hallucination, high computational costs, and fine-tuning difficulties highlighted areas for future improvement.

6. Bibliography

References

[1] Hongyang (Bruce) Yang1 , Xiao-Yang Liu1 , Christina Dan Wang, "FinGPT: Open-Source Financial Large Language Models" [Online]. Available: https://arxiv.org/abs/2306.06031

[2] Shijie Wu, Ozan Irsoy, Steven Lu, Vadim Dabravolski, Mark Dredze, Sebastian Gehrmann, Prabhanjan Kambadur, David Rosenberg, Gideon Mann"BloombergGPT: A Large Language Model for Finance"2023[Online]. Available: https://arXiv:2303.17564

[3] Teodoro Baldazzi, Luigi Bellomarini, Stefano Ceri, Andrea Colombo, Andrea Gentili, Emanuel Sallinger "Fine-tuning Large Enterprise Language Models via Ontological Reasoning," 2023. Available:https://arxiv.org/abs/2306.10723

[4] Zhihong Shao, Yeyun Gong, Yelong Shen, Minlie Huang, Nan Duan, Weizhu Chen, "Enhancing Retrieval-Augmented Large Language Models with Iterative Retrieval-Generation Synergy," https://arxiv.org/abs/2305.15294, 2023.

[5] [Bao et al., 2021] Siqi Bao, Huang He, Fan Wang, Hua Wu, Haifeng Wang, Wenquan Wu, Zhihua Wu, Zhen Guo, Hua Lu, Xinxian Huang, et al. Plato-xl: Exploring the large scale pre-training of dialogue generation. arXiv preprint arXiv:2109.09519, 2021.

[6] [Brown et al., 2020] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. Advances in Neural Information Processing Systems, 33:1877–1901, 2020.

[7] [Radford et al., 2018] Alec Radford, Karthik Narasimhan, Tim Salimans, Ilya Sutskever, et al. Improving language understanding by generative pre-training. OpenAI, 2018.

[8]Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, Denny Zhou, "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models". 2022. Available:https://arxiv.org/abs/2201.11903

[9][Thoppilan et al., 2022] Romal Thoppilan, Daniel De Freitas, Jamie Hall, Noam Shazeer, Apoorv Kulshreshtha, Heng-Tze Cheng, Alicia Jin, Taylor Bos, Leslie Baker, Yu Du, et al. Lamda: Language models for dialog applications. arXiv preprint arXiv:2201.08239, 2022.

[10][Vaswani et al., 2017] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Ł ukasz Kaiser, and Illia Polosukhin. Attention is all you need. In I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, Advances in Neural Information Processing Systems, volume 30. Curran Associates, Inc., 2017.

[11][Wu et al., 2023] Shijie Wu, Ozan Irsoy, Steven Lu, Vadim Dabravolski, Mark Dredze, Sebastian Gehrmann, Prabhanjan Kambadur, David Rosenberg, and Gideon Mann. BloombergGPT: A large language model for finance. arXiv preprint arXiv:2303.17564, 2023.

[12][Devlin et al., 2018] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:1810.04805, 2018.

[13][Dredze et al., 2016] Mark Dredze, Prabhanjan Kambadur, Gary Kazantsev, Gideon Mann, and Miles Osborne. How twitter is changing the nature of financial news discovery. In proceedings of the second international workshop on data science for macro-modeling, pages 1–5, 2016.

[14][Ethayarajh, 2019] Kawin Ethayarajh. How contextual are contextualized word representations? comparing the geometry of bert, elmo, and gpt-2 embeddings. arXiv preprint arXiv:1909.00512, 2019.

[15][Hu et al., 2021] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. International Conference on Learning Representations, 2021.

[16][Lewis et al., 2019] Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Ves Stoyanov, and Luke Zettlemoyer. Bart: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension. arXiv preprint arXiv:1910.13461, 2019.

[17][Lewis et al., 2020] Patrick Lewis, Myle Ott, Jingfei Du, and Veselin Stoyanov. Pretrained language models for biomedical and clinical tasks: understanding and extending the state-of-the-art. In Proceedings of the 3rd Clinical Natural Language Processing Workshop, pages 146–157, 2020.

7. My Appendix

List of Abbreviations and Acronyms

- **AI**: Artificial Intelligence
- **BERT**: Bidirectional Encoder Representations from Transformers
- **GPT**: Generative Pre-trained Transformer
- **LLM**: Large Language Model
- **NLP**: Natural Language Processing
- **RAG**: Retrieval-Augmented Generation
- **RLHF**: Reinforcement Learning from Human Feedback
- **SQL**: Structured Query Language
- **GPU**: Graphics Processing Unit
- **DAVID**: Database for Annotation, Visualization and Integrated Discover

CRediT authorship contribution statement

Nashit Budhwani: Conceptualization of this study, Methodology, Software, Original draft preparation. **Fahad Zahid:** Conceptualization of this study, Methodology, Software, Original draft preparation. **Lohit Ashawa:** Concep- tualization of this study, Methodology, Software, Original draft preparation. **Muhmmad Rafi:** Conceptualization of this study, Review of Research.

Nashit Budhwani earned his Bachelor's degree in Computer Science from the National University of Computer and Emerging Sciences (FAST) in 2024. His academic and research interests focus on artificial intelligence, machine learning, and data science, with a strong inclination toward AI-driven trade intelligence and automation. He played a pivotal role in TradeSage, a project aimed at transforming raw trade data into structured insights using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and vector-based indexing. His contributions involved fine-tuning AI models, optimizing data retrieval mechanisms, and enhancing trade analytics using NLP techniques. Passionate about leveraging AI and computational intelligence for real-world applications, Nashit continues to explore cutting-edge AI methodologies, aspiring to develop impactful, data-driven solutions for global challenges.

Fahad Zahid earned his Bachelor's degree in Artificial Intelligence from the National University of Computer and Emerging Sciences (FAST) in 2024. His academic and research interests lie in machine learning, natural language processing, and data engineering, with a strong focus on AI-driven trade analytics and financial intelligence systems. He was a key contributor to TradeSage, where he applied Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and vector-based indexing to enhance trade foresight and automate data-driven decision-making. His work involved fine-tuning AI models, structuring trade data for model training, and optimizing information retrieval for financial applications. Passionate about harnessing AI for real-world impact, Fahad continues to explore advanced AI methodologies and scalable data

solutions, aspiring to drive innovation in trade intelligence and enterprise AI systems.

Lohit Ashwa earned his Bachelor's degree in Artificial Intelligence from the National University of Computer and Emerging Sciences (FAST) in 2024. His academic and research interests lie in machine learning, natural language processing, and data-driven decision-making, with a focus on AI-powered trade analysis and intelligent automation. He was a key contributor to TradeSage, where he applied Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and vector-based indexing to optimize trade foresight and enhance data-driven insights. His work involved fine-tuning AI models, implementing scalable retrieval mechanisms, and improving financial analytics using NLP techniques. Passionate about AI-driven solutions, Lohit continues to explore cutting-edge methodologies in deep learning, reinforcement learning, and knowledge representation, aiming to build scalable and impactful AI applications.